

CODE: GS-CS

M.Tech. Common Entrance Test, PGCET – 2009

Computer Science and Engineering

Time: 2 Hours

Max. Marks: 100

Read the following instructions before answering the test

- i) Write / Shade the particulars of your identity, Test Seat Number and put your signature on the OMR Response Sheet before the start of the test.
- ii) All Questions have multiple choices of answers, of which only one is correct.
- iii) Mark the correct answer by completely shading only one circle against the Question number using Black Ink Ball Point pen only.
- iv) There will be no negative evaluation with regard to wrong answers. Marks will not be awarded if multiple answers are given.
- v) Do not make any stray mark on the OMR Response sheet. For rough work, use blank page on the question paper.
- vi) Taking the question paper out of the test hall is permitted only after the full duration of the test.
- vii) Use of only non-programmable calculator is permitted.
- viii) **START ANSWERING ONLY AT THE SPECIFIED TIME WHEN THE INVIGILATOR GIVES INSTRUCTIONS.**

MARKS DISTRIBUTION

PART – I	50 Questions :	$50 \times 1 =$	50 Marks
PART – II	25 Questions :	$25 \times 2 =$	50 Marks
		Total =	100 Marks

PART-I

Each Question carries One mark.

50 X 1 = 50 Marks

1. The density functions for X and Y are obtained from the joint density function in a manner analogous to the discrete case are called
a) Marginal b) Covariance c) Conditional d) Correlation
2. If the total number of items is Odd and Equal to say n then, the value of $\frac{1}{2}(n+1)^{\text{th}}$ gives:
a) The Mean b) The Median c) The Mode d) Harmonic Mean
3. The Harmonic Mean of 4,8,6 is
a) 8.853 b) 8.583 c) 5.853 d) 6.853
4. The Square of the Standard Deviation, σ^2 is called
a) Mean b) Median c) Variance d) Average Deviation $\frac{3}{\frac{1}{4} + \frac{1}{8} + \frac{1}{6}}$
5. If the terms are clearly the successive terms in the expansion $(q+p)^n$ then the distribution is called
a) Normal b) Binomial c) Poisson d) Gaussian $V = \sigma^2$
6. If $n(A) = n$, then $P(A)$ will contain
a) 2^n elements b) 2^{n-1} elements c) 2^p elements d) n^n elements $V \vee V$ laxman
normal successive terms
7. If A & B are disjoint, then $n(A \cup B)$ is equal to
a) $n(A) + n(B)$ b) $n(A) - n(B)$ c) $n(A) \cdot n(B)$ d) $n(A \times B)$ $A = \{1, 2, 3\}$ $|A| = 2^3 = 8$
8. Suppose $f: A \rightarrow B$ be a function such that $f(x) = x, \forall x \in A$ then f is called as
a) Constant function b) Polynomial function $A = \{1, 2\}$ $B = \{3, 4\}$
c) Identity function d) Relational function $A \cup B = n(A) + n(B)$ $1, 2, 3$
 $1, 2, 2$
9. $C_{n-1} + C_n$ is equal to
a) $2C_n$ b) $C_{n/2}$ c) $C_n \frac{(n+1)}{2}$ d) $C_n \frac{(n-1)}{2}$ *identity*
mirror image $x \cdot 1 = x$
 $2 \cdot 1 + 0 = x$
10. Which one of the following algorithm design technique used in finding all pairs of shortest distances in graph?
a) Dynamic Programming b) Backtracking c) Greedy d) Divide & Conquer
11. The Time Complexity of the Merge Sort is :
a) $O(\log n)$ b) $O(n)$ c) $O(n \log n)$ d) $O(n^2)$
12. The best data structure to check whether an arithmetic expression has balanced parenthesis is :
a) Queue b) Stack c) Tree d) List
13. The tightest lower bound on the number of comparisons in worst case for comparison based sorting is of the order of
a) n b) n^2 c) $n \log n$ d) $n \log^2 n$
14. An $m \times n$ matrix A is said to be sparse if:
a) many of its elements are non-zero b) many of its elements are zero
c) many of its elements are unit d) many of its elements are negative

15. The node of the linked list contains at least two fields :

- a) One for structure part & one for next pointer
- b) One for information part & one for structure part
- c) One for information part & one for next pointer
- d) One for pointer part & one for the index of the array

16. Each operator in a postfix string refers to the :

- a) Next operand in the string
- b) Previous operand in the string
- c) Next two operand in the string
- d) Previous two operands in the string

17. Following algorithm performs:

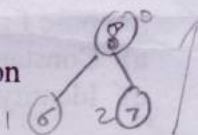
```
{
  for ( i=n; i>1; i--)
  {
    t = a[i];
    a[i] = a[1];
    a[1] = t;
    adjust (a,1,-n);
  }
}
```

RE/RL FA
CFG PDA
REL TM

- a) Merge sort
- b) Quick sort
- c) Bubble sort
- d) Heap sort

18. If L is L(A) for some DFA A, then L is said to be:

- a) Transition function
- b) Extended Transition function
- c) Regular Language
- d) Irregular Language



19. If 'w' is of length 'n', & the NFA has 's' states, then the running time of this algorithm is:

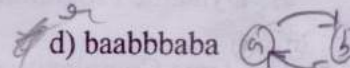
- a) $O(ns^2)$
- b) $O(ns)$
- c) $O(s^2)$
- d) $O(s)$

FA $\rightarrow$ RL
RE

20. The function h given by $h(0) = abb$ and $h(1) = ba$ is a homomorphism, then $h(1011)$ is equal to:

- a) abababab
- b) babababa
- c) abbbabaabb
- d) baabbbaba

NSF $\rightarrow$ NFA



21. Let $P = (Q, \Sigma, \Gamma, \delta, q_0, z_0)$ be a PDA. Then there is a context-free grammar G, such that:

- a) $L(G) = N(Q)$
- b) $L(G) = N(P)$
- c) $L(G) = N(G)$
- d) $L(G) \neq N(Q)$

$h(0) = ba$
 $h(1) = ba$
 $h(0) = ba$
 $h(1) = ba$

22. If $L = N(P)$ for some DPDA P, then L has:

- a) An ambiguous context-free grammar
- b) An unambiguous context-free grammar
- c) Inherently ambiguous context-free grammar
- d) Both a & c

$P = G$

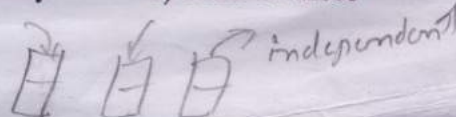
23. The set of languages we can accept using a Turing Machine is often called:

- a) Context-free languages
- b) Non-recursively enumerable languages (NRE-Languages)
- c) Recursively enumerable languages (RE-Languages)
- d) Machine Languages

deterministic = sure

24. In Multiple Turing machines, the heads move:

- a) Independently
- b) Dependently
- c) Sequentially
- d) None of these



25. Given relations:
 ➤ Employee (Name, Salary, Deptno)
 ➤ Department (Deptno, Deptname, Address)
 Which of the following queries cannot be expressed using basic relational algebra operations?
 a) Department address of every employee
 b) Employees whose name is same as their department number
 c) The sum of all employees salaries
 d) All employees of given department
26. Which Normal form is considered adequate for normal relational database design:
 a) 2NF b) 5NF c) 4NF d) 3NF
Name Deptname
27. Which of the following scenarios may lead to an irrecoverable error in a database system?
 a) A transaction writes a data item after it is read by an uncommitted transaction
 b) A transaction reads a data item after it is read by an uncommitted transaction
 c) A transaction reads a data item after it is written by a committed transaction
 d) A transaction reads a data item after it is written uncommitted transaction
primary key
28. In relational algebra operations, which of the following statements is /are true?
 I. UNION is commutative operation
 II. INTERSECTION is commutative operation
 III. SET DIFFERENCE is commutative operation
 a) I only b) II c) III d) I and II
*w commut ✓
 error*
29. Minimize the following Boolean expression :
 $bc + \bar{a}bc + a\bar{b}c$
 a) $ab + \bar{a}c$ b) $\bar{a}\bar{b} + \bar{a}c$ c) $ab + \bar{a}c + \bar{a}bc$ d) $\bar{a}\bar{b} + \bar{a}b$
*AUB=BOA
 A∩B=B∩A
 A-B ≠ B-A*
30. Double - Dabble method is used to convert:
 a) Binary number to decimal number b) Decimal number to Binary number
 c) Octal number to Hexa-decimal number d) Hexa-decimal number to Octal number
31. With 8-bit signed arithmetic, all magnitudes must be between:
 a) 0 and 255 b) 0 and 127 c) 0 and 256 d) 0 and 128
Double Decimal
32. The gated or clocked RS and D-flip flops are
 a) Completely Transparent b) Semitransparent
 c) Translucent d) None of these
Dabble binary
33. The PRESET and CLEAR inputs of a flip flop are called:
 a) Synchronous inputs b) Asynchronous inputs
 c) Symmetrical inputs d) Asymmetrical inputs
*RS } $\frac{1}{2} = 50\%$
 D }*
34. The ' ' string in html is used to insert a :
 a) Decimal point b) Percentage sign c) Space d) Copyright Symbol
nbsp - space
35. Which of the following Web Server is capable of running ASP Scripts?
 a) Java Web Server b) Apache Web Server
 c) Internet Information Server d) Tomcat Web Server

36. What is a 'Cookie' in Web based terminology?
 a) Name - Value pair that web server sends to client machine
 b) Piece of information that client keeps on the server
 c) Text files kept on the Gateway Interface
 d) Bug in a CGI script
37. The 'Request' object of ASP is used to:
 a) Extract Server Information on Client
 b) Extract Client Information on Server
 c) Send Client Information to Server
 d) Send Server Information to the Client
38. The 'a' tag in html stands for :
 a) Horizontal Rule
 b) Anchor
 c) Adjust
 d) Paragraph
39. TTL shift register counters operate up to frequencies of :
 a) 1 to 10 MHz
 b) 10 to 15 MHz
 c) 20 to 30 MHz
 d) 1 to 10 GHz
40. Mutual Exclusion ensures:
 a) Synchronization
 b) Concurrency
 c) System Integrity
 d) Parallelism
41. The decision to add to the pool of processors to be executed is called:
 a) Long term scheduling
 b) Medium term scheduling
 c) Short term scheduling
 d) Processor scheduling
42. The Address generated by the CPU is commonly referred to as a:
 a) Physical address space
 b) None
 c) Logical address space
 d) Virtual address space
43. A piece of core that is executed in response to an external event is called:
 a) a subroutine
 b) a program module
 c) an Interrupt service routine
 d) an application routine
44. The simplest ARQ scheme is the:
 a) Sliding Window Algorithm
 b) Transmitter Access Protocol
 c) Stop & Wait Algorithm
 d) None
45. Resilient Packet Ring is a:
 a) Standardized IEEE 802.05
 b) Standardized IEEE 802.11
 c) Standardized IEEE 802.16
 d) Standardized IEEE 802.17
46. Ethernet and Token Rings implement:
 a) Unicast in hardware
 b) Multicast in hardware
 c) Broadcast in hardware
 d) None
47. IPv6 Protocol is a :
 a) Unicast listener discovery
 b) Multicast listener discovery
 c) Broadcast listener discovery
 d) Internet Group Management Protocol
48. A top down parsing method is known as:
 a) Recursive descent
 b) Non-Recursive descent
 c) Semantic Routine
 d) Token specifier
49. Which of the Analysis involves in scanning the program to be compiled and recognizing the tokens?
 a) Syntactic Analysis
 b) Lexical Analysis
 c) Semantic Analysis
 d) Parser Analysis
50. Overlapping the execution of successive instructions is called:
 a) Queuing
 b) Pipelining
 c) Multitasking
 d) Non-Pipelining

PART-II

Each Question carries Two marks.

25 X 2 = 50 Marks

- ✓ 51. For all children taking an examination, the mean mark was 60% with standard deviation of 8%. A particular class of 30 children achieved an average of 63%. Is this unusual?

a) $P(\bar{X} > 63) = 0.020$
 c) $P(\bar{X} > 63) = 0.032$

b) $P(\bar{X} < 63) = 0.020$
 d) $P(\bar{X} > 63) = 0.035$

- ✓ 52. The Mean and Variance of the Poisson distribution with parameter λ are both equal to

a) λ, λ^2

b) λ^2, λ^2

c) $\lambda, 2\lambda$

d) λ, λ

- ✓ 53. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 3x + 1$, then f is a

a) Single valued function

b) Radical function

c) Bijective function

d) Relative function

54. In S_4 , if $f = \begin{pmatrix} a & b & c & d \\ b & c & d & a \end{pmatrix}$ and $g = \begin{pmatrix} a & b & c & d \\ b & d & c & a \end{pmatrix}$ then $f \circ g^{-1} =$

a) $\begin{pmatrix} a & b & c & d \\ a & b & d & c \end{pmatrix}$

b) $\begin{pmatrix} a & d & c & b \\ a & c & d & b \end{pmatrix}$

c) $\begin{pmatrix} b & c & d & a \\ a & b & d & c \end{pmatrix}$

d) $\begin{pmatrix} a & c & d & b \\ a & d & b & c \end{pmatrix}$

55. In Group $(\mathbb{Z}, *)$ where $a * b = a + b + 9$ then the identity element is:

a) $+9$

b) 9^{-1}

c) -9

d) -9^{-1}

56. The Context-free language for Palindromes $L(G_{pal})$, where G_{pal} is the grammar for palindromes, shown in fig

1	$P \rightarrow \epsilon$
2	$P \rightarrow 0$
3	$P \rightarrow 1$
4	$P \rightarrow 0P0$
5	$P \rightarrow 1P1$

is the set of palindromes over:

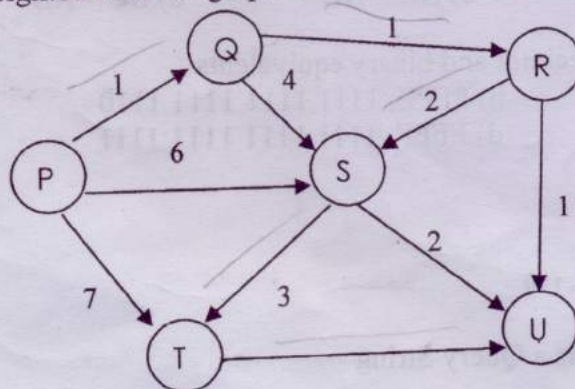
a) $\{0, 0\}$

b) $\{0, P\}$

c) $\{P, 0\}$

d) $\{0, 1\}$

57. Suppose we run Dijkstra's single source shortest path algorithm as the following edge-weighted directed graph with vertex P as the source



In what order do the nodes get included into the set of vertices for which the shortest path distances are finalized:

a) P, Q, R, S, T, U

b) P, Q, R, U, S, T

c) P, Q, R, U, T, S

d) P, Q, T, R, U, S

58. Consider the function f defined below :

```
struct item {  
    int data;  
    struct item *next;  
};  
int f(struct item *p) {  
    return ((p == NULL) || (p->next == NULL) || ((p->data <= p->next->data) &&  
        f(p->next)));  
}
```

For a given linked list p, the function f returns 1 if and only if:

- a) The list is empty or has exactly one element
b) The elements in the list are sorted in non-decreasing order of data value
c) The elements in the list are sorted in non-increasing order of data value
d) Not all elements in the list have the same data value

59. Assume that the operators +, -, x are left associative & Λ is right associative. The order of precedence (from highest to lowest) is Λ , x, +, -. The postfix expression corresponding to the infix expression $a + b \times c - d \Lambda e \Lambda f$ is :

- a) $abcx+def\Lambda\Lambda$ - b) $abcx+def\Lambda f\Lambda$ - c) $ab+cx-d-e\Lambda f\Lambda$ d) $-+axbc\Lambda\Lambda def$

60. The relation schema Student_Performance (name, courseno, rollno, grade) has the following dependencies

name, courseno $\rightarrow$ grade

rollno, courseno $\rightarrow$ grade

name $\rightarrow$ rollno

rollno $\rightarrow$ name

The highest normal form of their relation scheme is

- a) 2NF b) 3NF c) BCNF d) 4NF

61. A TTL circuit is said to provide 5MHz clock frequency with a stability better than 5 parts per million (ppm) over a 24-h time period. What are the frequency limits of the clock?

- a) 4,999,999 & 5,000,001 Hz b) 4,999,775 & 5,000,225 Hz
c) 4,999,975 & 5,000,025 Hz d) 4,999,995 & 5,000,005 Hz

62. Using Quine Mcclusky method, obtain the set of prime-implications for the function: $f(a,b,c) = \Sigma(2,6,7)$

- a) $\overline{B}\overline{C} + AB$ b) $\overline{B}\overline{C} + AB$ c) $BC + \overline{A}\overline{B}$ d) $BC + \overline{A}\overline{B}$

63. Convert decimal 65,535 to its Hexadecimal and binary equivalents

- a) FEFE, 1111 1110 1111 1110 b) FFFE, 1111 1111 1111 1110
c) FFF0, 1111 1111 1111 0000 d) FFFF, 1111 1111 1111 1111

64. What is a Query String in the URL :

- a) Text following the Colon (:)
b) Text following the Question Mark (?)
c) Text following the Hash (#)
d) The complete URL is referred to as a Query String

65. The $f_{\max}$ of a synchronous counter with series carry is grossly dependent on

- a) Propagation delay of AND gate and the number of flip flops in the counter
b) Propagation delay of flip flop and the number of flip flops in the counter
c) Propagation delay of flip flop and the propagation delay of AND gate
d) The number of flip flops in the counter and the propagation delay of AND gate

66. Consider a point-to-point link 50 kms in length. At what bandwidth would propagation delay (at a speed of 2×10^8 m/sec) equal transmit delay for 100 packets? What about 512 byte packets?
- a) 3.2 MB/sec, 16.4 MB/sec b) 2.2 MB/sec, 10.2 MB/sec
c) 1.6 MB/sec, 8.2 MB/sec d) 6.4 MB/sec, 64.8 MB/sec
67. Suppose a 128kbps point-to-point link is setup between Earth and a Rover on Mars. The distance from Earth to Mars is approximately 55Gm, data travels over the link at the speed of light. What is the minimum RTT for the link?
- a) 268 seconds b) 468 seconds c) 568 seconds d) 368 seconds
68. Security Association assigns an ID number called as:
- a) Security Index number(SIN) b) Security ID(SID)
c) Internet Key ID(IKID) d) Security Parameters Index(SPI)
69. Fixed Diffie-Hellman relies on :
- a) Certificates b) Signatures c) Public and Private key d) Public Key
70. A Digest Encrypted with public key algorithm but using the private key is called a:
- a) Kerberos b) Certification
c) Public key authentication d) Digital Signatures
71. The Quotient after dividing 100010010 by 1101 is :
- a) 11011 b) 10101 c) 11000 d) 00101
72. Alerting the Processor by a Hardware Signal is called :
- a) Deadlock b) Interrupt c) Procedure call d) Exception
73. Simplest way to determine cache locations to store memory blocks is:
- a) Direct Mapping b) Associative Mapping
c) Set-Associative d) In-Direct Mapping
74. A Husband and Wife appeared an interview for two Vacancies for the same post the Probability of Husband's selection is $1/7$ and that of Wife's selection is $1/5$. What is the Probability that only 1 will be selected
- a) $1/35$ b) $24/35$ c) $3/35$ d) $2/7$
75. Using K-map, minimize the given Boolean expression:
 $Y = ABCD + AB\bar{C}D + ABC\bar{D} + A\bar{B}\bar{C}D$. The minimized expression is:
- a) ABCD b) ABD c) AC d) AD

*** WISH YOU GOOD LUCK ***

MTech - PGCET 2009					
Computer Science					
Q.No	Answers	Q.No	Answers	Q.No	Answers
1	A	31	B	61	C
2	B	32	B	62	C
3	D	33	B	63	D
4	C	34	C	64	B
5	B	35	C	65	A
6	A	36	A	66	A
7	A	37	B	67	D
8	C	38	B	68	D
9	C	39	C	69	A
10	A	40	C	70	D
11	C	41	A	71	D
12	B	42	C	72	B
13	B	43	C	73	A
14	B	44	C	74	D
15	C	45	D	75	C
16	D	46	B		
17	D	47	B		
18	C	48	A		
19	A	49	B		
20	D	50	B		
21	B	51	A		
22	B	52	D		
23	C	53	C		
24	A	54	A		
25	B	55	D		
26	D	56	D		
27	C	57	B		
28	D	58	B		
29	A	59	A		
30	B	60	A		